

Real-Time Automated Cyber Threat Classification and Emerging Threat Detection Framework

ALEMAYEHU TILAHUN HAILE ¹ (Graduate Student Member, IEEE), SURAFEL LEMMA ABEBE ²,
AND HENOCK MULUGETA MELAKU ¹

¹School of Information Technology and Engineering, College of Technology and Built Environment, Addis Ababa University, Addis Ababa 1176, Ethiopia

²School of Electrical and Computer Engineering, College of Technology and Built Environment, Addis Ababa University, Addis Ababa 1176, Ethiopia

CORRESPONDING AUTHOR: ALEMAYEHU TILAHUN HAILE (e-mail: alemayehuthaile1@gmail.com).

ABSTRACT Automating cyber threat intelligence (CTI) collection and analysis in real time is critical for the timely detection and mitigation of cyber threats. Cybersecurity researchers have recently recommended CTI as a proactive and robust method for automated cyber threat prediction. This automated solution collects and analyzes real-time data from social media, cybersecurity forums, and hacker forums where cybersecurity analysts and hackers discuss cybersecurity-related topics to discover potential threats. In this article, we propose a comprehensive framework that automates both cyber threat classification and emerging threat detection using real-time data from surface, deep, and dark web sources. We collected real-time data from hackers and security forums to construct binary and multiclass cyber threat classifications. We employed a labeled leaked dataset to be considered as ground truth for classification. Machine and deep learning techniques were used to perform the classification. Latent Dirichlet allocation (LDA) and nonnegative matrix factorization (NMF) were used to analyze topic distribution over time and identify emerging threats. This approach allows for the identification of zero-day attacks and other emerging threats by monitoring shifts in topics. Using a support vector machine with the bag-of-words (binary term weight) model achieved the highest accuracies of 93.67 and 96.35 for binary and multiclass classifications, respectively. Moreover, LDA and NMF were used to extract the top topics from various numbers of topics. The LDA model is well suited for identifying emerging trends and useful for real-time threat monitoring in cybersecurity.

INDEX TERMS Cyber threat intelligence (CTI), emerging threat detection, Latent Dirichlet allocation (LDA), natural language processing, nonnegative matrix factorization (NMF), threat categorization, zero-day attacks.

I. INTRODUCTION

As cyber threats have become more sophisticated and persistent, organizations are becoming increasingly reliant on cyber threat intelligence (CTI) to detect and prevent cyber threats. CTI collects, analyzes, and disseminates cyber threat information; hence, CTI enables organizations to better understand the threat landscape and make informed cybersecurity-related decisions. The massive unstructured data used for CTI are mostly collected from online sources, such as web forums, social media platforms, blogs, news articles, and incident analyses. Hacker forums are the most valuable source for cyber threat-related posts and blogs, which usually contain

essential information about possible cyber threats or vulnerabilities [1]. However, cybersecurity analysts experience significant challenges when extracting and analyzing CTI, as manually processing this information is time consuming and prone to human error. To address these challenges and extract actionable CTI, automated solutions using machine learning (ML), natural language processing (NLP), and topic modeling are essential [2].

Threat classification is the process of organizing collected and analyzed CTI data into specific classes, such as malware, phishing, denial-of-service (DDoS) attacks, or ransomware. Accurate classification allows cybersecurity teams to quickly

identify the nature of a threat and prioritize their responses based on severity or relevance to their organization. The traditional manual method requires vast amounts of textual data to be processed before cybersecurity analysts can identify threats. Manual approaches are inefficient due to the complexity and volume of CTI data; therefore, they usually fail to capture all relevant information [3]. The use of ML and NLP models to automate threat classification addresses these limitations by processing large-scale unstructured data and identifying key features indicative of diverse types of cyber threats [4]. This process ensures that cyber threats are classified consistently and in real time. Hence, automated real-time classification significantly improves cyber threat response times and helps organizations stay ahead of their adversaries by identifying and responding to cyber threats more efficiently [5].

The ability to detect emerging threats, such as zero-day vulnerabilities, is vital for the success of modern cybersecurity strategies. Zero-day attacks are risky because they exploit unknown vulnerabilities in software, leaving systems exposed until a patch is developed. Early detection of these threats is crucial to preventing widespread damage. Topic modeling techniques, such as latent Dirichlet allocation (LDA) and nonnegative matrix factorization (NMF), can be employed to identify emerging threats by uncovering hidden patterns and shifts in threat vectors within CTI data over time [6].

The probabilistic LDA model assumes that every document contains a variety of themes based on word combinations, which makes it possible to extract underlying patterns from CTI data. These patterns may then be monitored over time to determine whether new threat vectors are emerging or previously known [7]. Organizations can identify early warning indicators for emerging threats by examining the temporal distribution of topics, which gives them an opportunity to reduce their risks before these threats become widespread attacks. Security analysts can proactively act defensively by keeping an eye on changes in the discussion frequency of specific topics, such as a sudden increase in discussions about a new ransomware variant, which can indicate the appearance of a new threat. Similarly, the NMF model is effective at identifying part-based representations of data, which makes it ideal for identifying specific components of cyber threats, such as adversarial tactics, techniques, and procedures (TTPs) [8]. Integrating the LDA and NMF models into emerging threat detection not only enhances cybersecurity analysts' ability to detect zero-day attacks but also provides a comprehensive view of the evolving cyber threat landscape. To the best of our knowledge, this is the first work that jointly performs real-time cyber threat classification and emerging threat detection using machine learning, deep learning, and topic modeling techniques in real-time from surface, deep, and dark web forums.

The two key contributions of this article are as follows:

- 1) Automated cyber threat classification introduces an ML- and NLP-based approach to classifying unstructured CTI data into predefined threat categories, such as malware, phishing, DDoS attacks, and ransomware.

This automated system helps reduce the manual effort required for threat classification and increases accuracy and consistency.

- 2) Emerging threat detection is based on LDA and NMF models to identify emerging threats using topic modeling over time, which enables the identification of zero-day attacks by monitoring the evolution of topics within CTI data.

These contributions not only improve the automation of cyber threat analyses but also provide practical tools for cybersecurity teams to enhance their threat detection and response strategies.

II. RELATED WORKS

The need for effective analysis of CTI data has led to the development of several approaches for automating threat detection, classification, and analysis. In this section, we review the literature related to threat classification and emerging threat detection using ML, NLP, and topic-modeling techniques.

A. THREAT CLASSIFICATION

The process of manually classifying cyber threats is time-consuming and mistake-prone, particularly considering the unstructured nature of the data found in CTI sources, such as articles, blogs, and reports. Researchers have suggested using automatic classification systems based on NLP and ML techniques to overcome these difficulties. In their survey of cybersecurity intelligence models, Irshad and Siddiqui [9] explored the expanding trend of using NLP and ML models in CTI classification. Their research highlighted the benefits of automating the classification of documents into different classes, such as malware, phishing, and DDoS attacks, which can improve the speed and accuracy of threat detection. Yang and Lam [10] employed ML to create an automated CTI classification system based on a multiclassification framework by combining two document-embedding models with six ML classifiers, which grouped similar threat reports together before analysis. Their experimental results demonstrated the applicability of the framework for efficiently processing a very large number of CTI reports for use by cybersecurity analysts.

Behzadan et al. [11] presented a framework for the identification and classification of cyber threat indicators in the Twitter stream. This framework offers a data-driven method for modeling and classifying tweets linked to cybersecurity events based on a cascading convolutional neural network (CNN) architecture using a binary classifier with a multi-class model to categorize cyber-related tweets into various cyber threats. Liu et al. [12] proposed a trigger-enhanced actionable CTI discovery system that uses NLP technology to produce actionable CTI from cybersecurity reports. This system illustrates the connections between the indicators of compromise (IoCs) and the campaign stages of attack vectors. To enhance the effectiveness of their classification model, Liu et al. [12] specifically introduced a "campaign trigger,"

which uses keywords to identify the various campaign stages efficiently. Dionísio et al. [13] employed deep neural networks to classify cyber threat indicators from Twitter, and supporting the viability of using social media and deep learning for real-time CTI extraction.

B. EMERGING THREAT DETECTION

Detecting new or evolving threats, such as zero-day attacks, is crucial for maintaining a proactive cybersecurity posture. Basheer et al. [14] underscores the growing relevance of dark web monitoring in cyber threat intelligence and highlighting key challenges in extracting and correlating threat signals from hidden services and forums. Traditional intrusion detection systems often fail to recognize novel cyber threats. To overcome this limitation, researchers have applied topic modeling techniques to uncover hidden patterns in CTI data. For example, LDA is one of the most widely used probabilistic topic modeling techniques in cybersecurity. Blei et al. [6] first introduced LDA as a generative model for discovering topics in large text corpora, and its application to CTI data has gained significant traction. Vishwanathan et al. [5] applied LDA to CTI datasets to detect latent threats by analyzing document topics and their temporal distributions. This approach enables the detection of shifts in threat vectors, which helps identify emerging threats that may not be detected by traditional methods.

Similarly, NMF has been explored as an alternative to LDA for emerging threat detection. NMF is a matrix factorization technique used to identify part-based representations of data. Li et al. [15] proposed an open-source intelligence framework for cybersecurity that can automatically analyze unstructured text to produce event-based CTI data. The framework uses a number of methods, such as ML, data mining, and NLP, to extract cybersecurity event-related data, common vulnerabilities, and exposure to profile threat actors. Kadoguchi et al. [16] proposed using various methodologies, such as ML and NLP, to extract intelligence from dark web forums, which allows cybersecurity analysts to identify emerging cyber threats and take appropriate measures against malicious activities. In particular, they applied multilayer perceptron deep learning (DL) using the feature values obtained by the document-to-vector (Doc2Vec) model for CTI classification.

Lau and Leung [17] proposed a hybrid method for threat detection and classification that combines DL, NMF, and LDA approaches to enhance their method's ability to predict new threats based on existing data, in addition to taking advantage of these approaches for topic modeling.

Suryotrisongko et al. [18] proposed topic modeling techniques for CTI applications after providing a quick overview of topic modeling applications. Suryotrisongko et al. [18] used BERTopic and Top2Vec techniques to allow users to choose their own pretrained text/sentence-embedding model, which supports several languages. In particular, the Open Web Application Security Project (OWASP) Maryam: Open-Source Intelligence (OSINT) framework was used for the proposed modeling module. In addition, Suryotrisongko et al. [18]

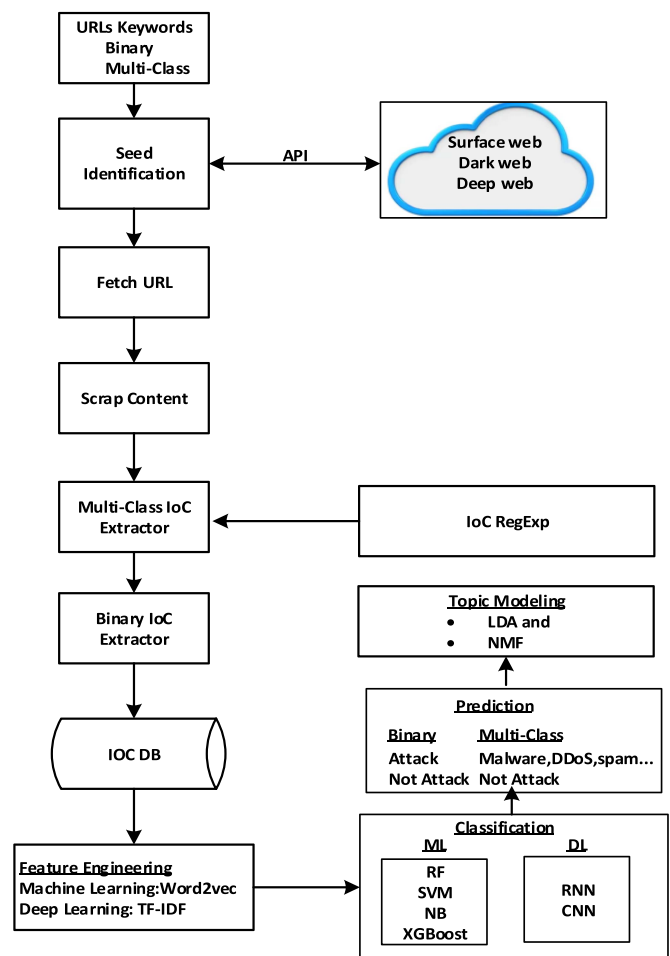


FIGURE 1. Proposed framework for cyber threat intelligence collection, classification, and emerging threat detection.

used data from a hacker forum leak to validate experiment outcomes.

C. GAPS IN THE LITERATURE

Despite the significant literature on threat classification and emerging threat detection using ML, NLP, and topic modeling, few studies have considered threat classification and emerging threat detection together. Most earlier approaches focused solely on detecting threats based on static datasets, which limits their ability to track how threats evolve over time. Furthermore, to the best of our knowledge, no research has considered real-time threat classification and emerging threat detection based on the surface web, deep web, and dark web.

III. PROPOSED FRAMEWORK

This section presents the proposed framework illustrated in Fig. 1, which outlines the full pipeline from data acquisition to classification and emerging threat analysis. We incorporate approaches adapted from prior CTI studies such as Irshad and Siddiqui et al. [9], Yang and Lam et al. [10], and Li et al. [15], which demonstrated automated intelligence extraction and classification from unstructured online sources.

A. DATA COLLECTION

We collected CTI data shared by cybersecurity agents on the surface web, deep web, and dark web. These agents are usually real hackers, analysts, experts, and individuals who are passionate about cybersecurity threat protection. However, collecting CTI data from these sources presents several challenges. For example, many sources restrict access to their content to registered users, which makes it more difficult for web crawlers to perform real-time automated data collection. Other sources may not only require registration but also a fee-based subscription to get API access. Furthermore, some sites may use CAPTCHAs to stop automated programs from collecting data, while others have invitation-only registration systems that require a current member to invite new users to register. In this study, we used a source that did not have any of these restrictions. In addition, we focused on sources with a large amount of URL data about CTI. We also paid a subscription fee to obtain API access. We used Python to create web crawlers that collect information from these sources. The purpose of these crawlers was to navigate various URLs and retrieve unstructured content. When feasible, we used registered accounts to manually collect data from sources with access limitations and otherwise employed a number of steps to manually intervene and get around CAPTCHAs during data collection. Using cybersecurity-related keywords, for example, we found URLs that were likely to have useful feeds pertaining to CTI. Our Python web crawlers were used to handle these different source structures and retrieve pertinent content. Moreover, we used a publicly accessible leaked dataset previously labeled by experts as a ground-truth dataset to assess the effectiveness of the trained binary and multiclass classification models (see nulled.io).

All the data collected were structured, and their contents were labeled according to relevance and threat category. Regular expressions were used to extract the IoCs from the text, including domain names, email addresses, IP addresses, and URLs. Finally, we combined all the gathered information into a complete dataset that featured distinct threat categories for binary and multiclass classifications. To produce actionable CTI, this dataset served as the foundation for further analysis and classification operations.

B. DATA PREPROCESSING

Data collected from the surface web, deep web, and dark web are typically unstructured and noisy; therefore, the data must be preprocessed comprehensively before constructing the final CTI datasets. The first step in the preprocessing pipeline is to clean up the HTML tags and extract the title and content of each post. To preserve consistency, we changed all text to lowercase after cleaning up the HTML. Special characters, punctuation, and stop words were removed to minimize noise. Next, we used lemmatization to break down words into their most basic forms. Lemmatization is a crucial step since it simplifies and optimizes the textual data for analysis by reducing inflectional and derivatively related word forms to a common

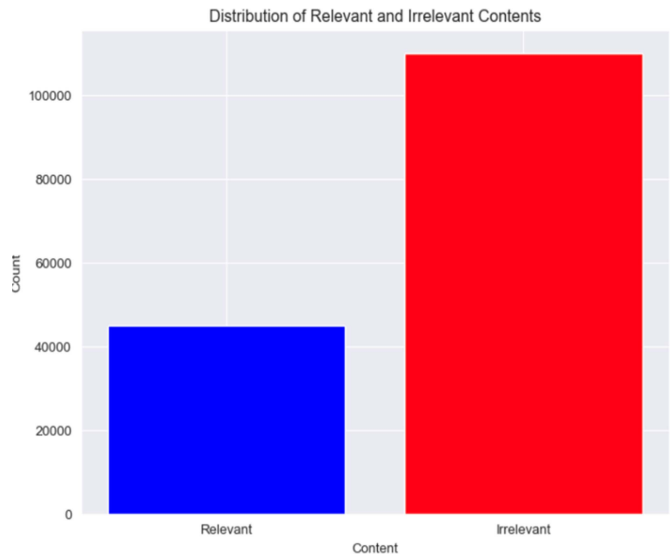


FIGURE 2. Post distribution according to relevant and irrelevant classes.

base form. Once the text was cleaned and normalized, each post was converted into a single-line document. Representing each post as a single line ensured that the text was formatted consistently, making it easier to process with text analysis tools. We aimed to build two separate datasets for binary and multiclass classifications using crawled and preprocessed data from cybersecurity forums, which were constructed using a keyword search method that labels each sample based on keywords.

C. BINARY AND MULTICLASS DATASET CONSTRUCTION

To construct a binary classification dataset, relevant and irrelevant labels were applied to the data. Posts containing cybersecurity-pertinent keywords were labeled as relevant. For example, the relevant terms “cyber intelligence feeds,” “cyber intelligence URLs,” “IoC feeds,” “malware domains list,” “phishing URLs,” “ransomware URLs feed,” “cybersecurity threat feeds,” and “cyber threat indicators,” among others, were used frequently. These keywords were selected with care to cover a wide spectrum of cybersecurity-related topics and ensure thorough coverage of relevant posts.

Posts with irrelevant terms, such as “popular websites,” “most visited websites,” “top news websites,” “top social media websites,” and “popular blogs,” or those without any cybersecurity-related keywords were labeled accordingly. In the labeling process, a post is considered irrelevant if it has no relevant keywords and if it contains no keywords related to cybersecurity. As shown in Fig. 2, this binary distinction helped create a dataset with a clear separation between cybersecurity-related and nonrelated content to facilitate easier classification.

We constructed a multiclass dataset from the posts labeled as relevant in the binary dataset, which were categorized into six distinct threat categories defined by specific keywords associated with different types of cyber threats, allowing for

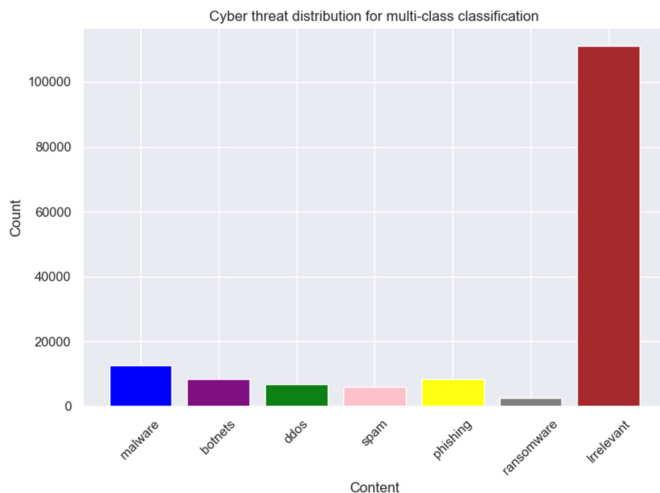


FIGURE 3. Post distribution for multiclass classification.

a more granular analysis of the data. Fig. 3 shows the post distribution for multiclass classification.

The categories and their associated keywords are as follows:

- *DDoS Attack*: Keywords included “DDoS,” “denial of service,” and “server.”
- *Botnets*: Keywords included “botnet,” “command and control,” and “C2.”
- *Malware*: Keywords included “malware,” “virus,” “worm,” and “trojan.”
- *Phishing*: Keywords included “phishing,” “scam,” and “fraud.”
- *Spam*: Keywords included “spam,” “junk mail,” and “unsolicited.”
- *Ransomware*: Keywords included “ransomware,” “encryption,” and “decrypt.”

We used the presence of these keywords to evaluate each post in the relevant class and to identify the most relevant category. This procedure made it possible to produce a multiclass dataset that could be used to train models for better threat detection and classification. We used these dataset-building procedures to properly structure and label our datasets, which is necessary for the efficient classification of CTI from the surface web, deep web, and dark web. This systematic approach to dataset construction allowed for accurate and meaningful classification, which ultimately helped in the identification of actionable CTI.

D. FEATURE ENGINEERING

ML algorithms cannot directly process textual data; instead, they require numerical input [16]. Therefore, before supplying documents to supervised or unsupervised learning models, they must be vectorized [19]. We applied basic and advanced feature engineering methods for ML and DL, respectively. We also used conventional feature engineering techniques to prepare our textual data for ML models. We started by applying the bag-of-words (BOW) model as a vector of terms where every dimension is associated with a distinct phrase from the

full corpus. We used two BOW approaches: term-frequency weights and binary term weights. The binary term weights approach uses a vector with values of 1 or 0 to indicate whether a term is present in a document. However, the term-frequency weights approach provides a more accurate representation of a document’s content. We employed term frequency-inverse document frequency (TF-IDF) to improve our feature representation. TF-IDF scores for individual words (i.e., unigrams) in the document were calculated by multiplying term frequency by inverse document frequency to help downscale common terms across many documents [20]. Additionally, we extended the TF-IDF calculation to include pairs of consecutive words (i.e., bigrams) to capture contextual and semantic meanings that might be missed by unigrams. We used these TF-IDF features to produce a robust numerical representation of our textual data for ML models.

Word2Vec is one of the advanced feature engineering approaches used in this study, as it helped us discover semantic relationships between words in the corpus [21]. Word2Vec represents each word as a vector in a continuous vector space by positioning semantically similar words close to each other. We used our dataset to train our own Word2Vec model with a feature size of 100 because there are not many pretrained Word2Vec models tailored to cybersecurity. This training produced 100-dimensional vectors for each word in the document corpus. However, since Word2Vec does not offer document-level vectors directly, we computed the document vectors by averaging the word vectors of all the words in each document, summing the vectors, and then averaging them to obtain the final document-level vector.

We also used another advanced feature engineering method, Doc2Vec, which is an extension of word-to-vector (Word2Vec), focused on learning document-level vectors [22], [23]. In particular, Doc2Vec generates a fixed-length vector for an entire document to capture its context and semantics. We trained a Doc2Vec model on our dataset to directly obtain document-level vectors and effectively capture the overall sentiment and thematic structure of various documents. Hence, our Doc2Vec model is suitable for complex text analysis.

Every document in the dataset underwent a number of data-processing stages to incorporate these feature engineering methods into our pipeline. Initially, we preprocessed the text by removing HTML tags, converting the text to lowercase, removing stop words and punctuation, and applying lemmatization. We then generated BOW vectors using both binary term weights and term-frequency weights for the ML models. To determine the relevance of terms in documents, we computed TF-IDF scores for bigrams and unigrams. We used Word2Vec on the preprocessed text to obtain word vectors, and calculated document vectors by averaging the word vectors for each document. Similarly, we used Doc2Vec on the preprocessed text to directly obtain document-level vectors for the ML models.

We ensured that the textual data were well represented using traditional and advanced feature engineering techniques,

which captured both the term frequency and semantic relationships. This robust feature engineering process established a foundation for effective ML and DL model training, which enables the accurate and meaningful analysis of CTI data from the surface web, deep web, and dark web.

E. CLASSIFICATION

The next step in our methodology is to classify textual data, identify relevant posts, and categorize them into specific types of cyber threats. We used both ML and DL models for binary and multiclass classification tasks. The main objective of binary classification is to automatically distinguish between relevant and irrelevant posts, while the objective of multiclass classification is to categorize relevant posts into six specific types of cyber threat (i.e., DDoS attacks, botnets, malware, phishing, spam, and ransomware). The first ML algorithm is the random forest (RF) classifier, an ensemble learning approach that constructs multiple decision trees during training and outputs the mode of the classes for classification tasks. The RF classifier is highly effective for handling high-dimensional data and provides excellent accuracy and robustness [24]. We used a support vector machine (SVM) algorithm as a supervised learning model to identify the hyperplane separating the classes in the feature space [25]. In particular, the linear version is suitable for high-dimensional textual data. We also used the multinomial naïve Bayes (NB) classifier, which is commonly known for its text classification effectiveness [26]. This probabilistic classifier works well with term-frequency features and provides strong baseline performance. Finally, we employed the eXtreme Gradient Boosting (XGBoost) package, which is an advanced boosting algorithm that combines the strengths of multiple weak models to enhance predictive power and is suitable for complex classification tasks [26]. In addition to these ML models, we employed DL models to capture more complex patterns in the textual data. For example, we built a CNN to capture local patterns [27]. This model was trained on word vectors obtained from the Word2Vec model. We also constructed a long short-term memory (LSTM) network to capture the sequential nature of the textual data, which consisted of embedding layers, LSTM layers with dropouts for regularization, and dense layers [28]. This model was trained on document vectors obtained from the Doc2Vec model.

These classification tasks help narrow down and focus only on the data that contain cybersecurity-related information (i.e., relevant posts). Irrelevant posts (such as general discussions about entertainment or other unrelated topics) were excluded from further processing to improve the quality of the data going into the topic modeling dataset.

F. EMERGING THREAT DETECTION

Traditional security solutions that rely on signature-based detection face serious problems from emerging cybersecurity threats, such as zero-day attacks, as these systems have trouble identifying previously unseen cyber threats. These problems

emphasize the need for an approach that can effectively recognize new attack patterns and cyber threats, such as topic modeling [4], which enables the automatic extraction of hidden patterns from unstructured data, such as blogs, event logs, and CTI reports. Organizations can detect changes in the cyber threat landscape, such as the emergence of new attack vectors or TTPs, by examining how these issues have changed over time. LDA and NMF are two popular techniques used for topic modeling [29], and their ability to uncover hidden structures in large datasets has been demonstrated in various domains, including cybersecurity [2].

The probabilistic LDA model assumes that a variety of themes can be found in every document, where each theme is defined by a word distribution [6]. This makes LDA well suited for detecting emerging trends, as topics gain or lose prominence in real time. In contrast, NMF breaks down the data into representations based on components and focuses on recognizing discrete and cohesive topics that frequently hold true throughout time. LDA is more sensitive to sudden changes in topic distribution, which makes it perfect for identifying recently emerging threats, while NMF is excellent at identifying long-term underlying trends in the data. These algorithms work together to effectively detect emerging threats, enabling cybersecurity analysts to keep an eye on both persistent and dynamic threats and ensure their strong defense against these evolving cyber threats.

IV. EXPERIMENTAL SETUP

We conducted the experiment on a machine running an Intel CPU at 2.20 GHz with 16 GB of RAM and an NVIDIA GeForce RTX 2070 GPU, and 64 GB of RAM running the Arch Linux operating system. We used standard regular expressions to perform data reprocessing. To determine the performance of the proposed approach, we developed a prototype using the Python 3.10 programming language in the Jupyter Notebook environment. We used the spaCy library to conduct lemmatization, while the scikit-learn libraries were used to develop the classifiers. The neural network-based classifiers were coded in Keras. We used both Gensim and scikit-learn libraries for topic modeling. We also used the Gensim library for training the Word2Vec and Doc2Vec models.

For each algorithm, we followed a structured approach to train and evaluate our models. First, we used training and testing datasets using an 80:20 ratio. Next, we used the appropriate hyperparameters to train each model on the training set. For the ML models, we tuned the hyperparameters using a grid search and cross-validation. For the DL models, we used early stopping and learning rate schedules to prevent overfitting. After training, we evaluated the models using the ground-truth dataset as a testing dataset. Performance metrics, such as accuracy, precision, recall, and F1-score, were used for evaluation, which enabled us to accurately identify and categorize CTI data from hacker forums, providing valuable insights for cybersecurity analysis. This classification task allowed us to focus only on relevant posts containing

TABLE 1. Classification Accuracy (%) Obtained by Classifiers for Binary Classification

	Classifiers	BOW (binary term weight)	BOW (term frequency weight)	TF-IDF (unigram)	TF-IDF (bigram)	Word2Vec	Doc2Vec
Machine learning	RF	89.88	89.89	88.3	69.01	64.64	73.85
	SVM	93.67	92.75	90.69	68.87	64.53	73.59
	NB	79.96	77.96	75.36	64.65	-	-
	XGBoost	91.4	90.3	88.3	70.45	58.4	63.19
Deep learning	CNN	87.00	90.02	86.1	72.9	91.22	86.44
	LSTM	88.9	90.5	88.3	67.03	93.13	88.90

TABLE 2. Classification Accuracy (%) Obtained by Classifiers for Multiclass Classification

	Classifiers	BOW (binary term weight)	BOW (term frequency weight)	TF-IDF (unigram)	TF-IDF (bigram)	Word2Vec	Doc2Vec
Machine learning	RF	89.25	94.14	91.86	70.03	53.54	63.19
	SVM	96.35	95.35	83.06	75.24	58.86	77.46
	NB	85.67	83.06	70.68	68.50	-	-
	XGBoost	92.30	91.13	86.00	72.5	54.53	66.10
Deep learning	CNN	88.98	92.34	83.81	78.83	90.11	84.10
	LSTM	90.3	93.9	83.06	64.5	94.01	87.72

cybersecurity-related information. After identifying relevant posts, these data were added to the topic modeling dataset to identify emerging cyber threats.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. CLASSIFICATION

The classification results described here validate the first major component of our proposed framework, namely the automated categorization of CTI using machine and deep learning models. The experimental results demonstrate the effectiveness of various classifiers across different feature extraction methods for binary and multiclass classification (see Tables 1 and 2, respectively). For example, SVM achieved high accuracy using the traditional feature extraction methods BOW and TF-IDF. In particular, SVM with BOW (binary term weight) achieved the highest accuracies of 93.67 and 96.35 for binary and multiclass classification, respectively. However, the binary term weights approach captures only the presence or absence of terms, which simplifies the feature space and tends to work well for models that rely on independence assumptions or high-dimensional sparse features.

XGBoost also performed well using BOW and TF-IDF, as XGBoost with BOW (binary term weight) achieved the highest accuracies of 91.40 and 92.30 for binary and multiclass classifications, respectively. In contrast, RF showed moderate performance with BOW and TF-IDF but did not support dense embeddings, such as Word2Vec and Doc2Vec, due to its reliance on discrete and frequency-based features.

The best performance when using NB was with TF-IDF in binary classification (0.84) and with BOW in multiclass classification (0.83). However, RF showed moderate

performance with BOW and TF-IDF. The best performance when using RF was with BOW (term frequency weight) in binary and multiclass classification (99.89 and 94.14, respectively). In addition, NB showed a notable decline in accuracy when applied to BOW and TF-IDF. These findings imply that both RF and NB are more suited to conventional feature extraction techniques than dense and continuous embeddings, which calls for models that can grasp semantic subtleties.

However, Word2Vec-based feature engineering techniques do not provide significant performance boosts. In most cases, specifically using multiclass datasets, Word2Vec-based feature engineering techniques generate results with less accuracy than techniques using BOW and TF-IDF weights. The reason for these less accurate results is the use of 100-dimensional vectors to represent the documents for both Word2Vec and Doc2Vec. In addition, the multiclass dataset had more categories and a smaller number of samples per class when compared to the binary datasets. Increasing the size of the feature vector may result in enhanced performance. However, training such a model will consume significant computing resources.

DL models, such as CNN and LSTM, have demonstrated notable benefits for dense embeddings such as Word2Vec and Doc2Vec. LSTM with Word2Vec had the best accuracy of 93.13 in the binary classification, followed by CNN with 91.22 accuracy. In a similar fashion, CNN showed an accuracy of 90.11 with Word2Vec, while LSTM outperformed the other models, achieving an accuracy of 94.01 for multiclass classification. This implies that the CNN and LSTM models are especially well suited to embeddings that maintain semantic information because of their capacity to recognize local

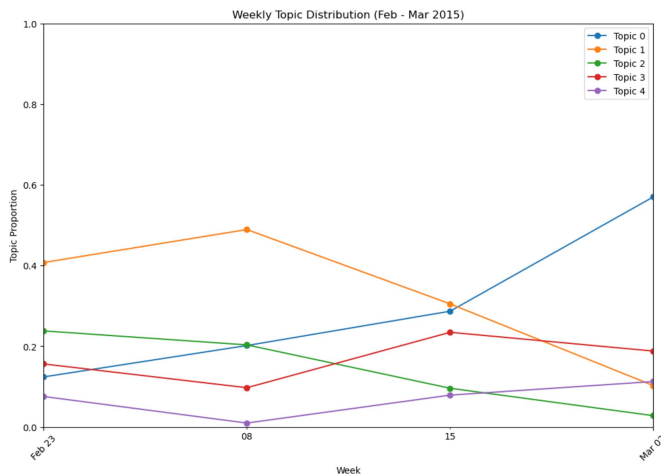


FIGURE 4. Weekly topic distribution using the LDA model (Feb.-Mar. 2015). *Topic 0:* pm, private, total, list, time, contact, forums, hack, details, posts. *Topic 1:* version, printable, subscribe, new, pm, quote, reply, view, thread, posts. *Topic 2:* thread, btc, average, prestige, joined, reputation, report, posts, post, pm. *Topic 3:* thread, xmpp, post, joined, reputation, prestige, report, quote, posts, pm. *Topic 4:* reputation, joined, prestige, report, service, posts, wrote, vouch, post, pm.

and sequential patterns, which enables them to distinguish between complex classes. Word2Vec, with its word-level context, is compatible with DL models, as demonstrated by the small decrease in accuracy observed by LSTM and CNN when using Doc2Vec embeddings, which capture document-level semantics but lack sequential word context. Overall, for binary and multiclass classification tasks, SVM with BOW (binary term weight) and LSTM with Word2Vec provided the highest accuracies among the ML and DL methods, respectively.

B. EMERGING THREAT DETECTION

This part of the results highlights the second emerging threat detection component of our proposed framework using LDA and NMF. In this experiment, we used LDA and NMF topic modeling techniques to examine the evolution of topics in the CTI data. The goal was to track changes in topics across a period of time. In other words, we had to determine the emergence of different threats or changes in direction within the scope of the CTI data. In this section, we present the outcomes of the LDA and NMF models regarding their topic distributions, and the significance of the subjects to CTI data analysis. Following the data analysis, we used the LDA model to identify five distinct topics in the corpus content, as represented by the most descriptive related words. These topics changed over several weeks; for example, the ratio of each topic changed between February 23 and March 2, 2015 (see Figs. 4 and 5).

A sharp increase in Topic 0 was seen in the second half of February and early March, which implies that there was an increase in discussions centered on various issues, including but not limited to private messages and forum scraping. This could indicate an increase in private messages, especially by

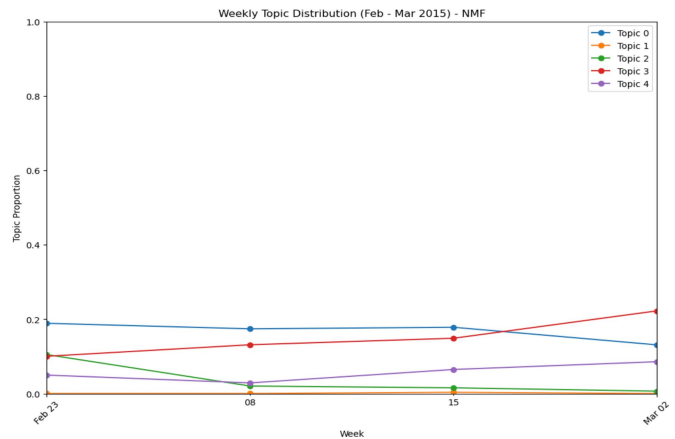


FIGURE 5. Weekly topic distribution using the NMF model (Feb.-Mar. 2015). *Topic 0:* xmpp, thread, quote, joined, prestige, reputation, report, post, posts, pm. *Topic 1:* subscribe, posts, emails, yahoo, og, closed, view, thread, pm, poopie. *Topic 2:* days, sort, ago, btc, thread, new, pm, vote, average, post. *Topic 3:* welcome, private, time, list, total, forums, contact, hack, details, posts. *Topic 4:* thanks, service, report, reputation, joined, prestige, post, vouch, wrote, pm.

threat actors coordinating their activities or by individuals sharing sensitive issues on the forums.

Notably, Topic 1 seems to indicate quotations and replies to quotations, among other activities, such as nominally engaging in threads with which other members are already interacting by creating new posts and threads. The word “version” points to some conversations touching on new types of attacks. Topic 1 shows downward trends throughout the timeline, indicating that the introduction of new threads, thread participants’ engagement, and their interactions through replies to threads reduced with time. This situation could imply that there was a high initial engagement with these new threads, which steadily fell as their content matured or as the related discussions shifted to more private forums (as captured in Topic 0).

Topic 2 is probably related to threads focused on Bitcoin (BTC), where the thread participants discuss BTC’s reputation, related posts, and general activities. The discussion of prestige and reputation in this topic may refer to the standing of forum users. In particular, Topic 2 remained relatively stable; therefore, it can be said that BTC-related activities and building user reputations did not dominate conversations during the examined period. However, users are involved in particular communication systems to a certain degree in addition to systems for enhancing reputations. This topic is similar to Topic 3, which is concerned with threads related to XMPP (a communication protocol for instant messaging) and users participating in forum activities.

Considering Topic 3, there is an observable slight upward trend, which denotes that the examined period witnessed increased forum discussions about topics related to instant messaging and reputation building. Hence, Topic 3 deals with the reputations of the forum users and the services that they provide, including vouching and reporting. In contrast, Topic

4 is still considered to be minor, with slight movements over time, which means that the issues of user rating and rating other user's services were given attention, but did not change over time as much as other issues.

The LDA model effectively tracked changes in the participants' conversations. For example, an increase in Topic 0 could signify a shift in relevance toward internal communications and the sharing of sensitive materials. That is, threat actors coordinate with each other or discuss their illicit activities. Conversely, the decrease in Topic 1 may indicate that the more general, open forum discussions were becoming decreasingly important as private discussions gained significance. This analysis demonstrates the capacity of the LDA model to detect emerging trends and monitor. Although increases in topic activity such as private messaging or threat-specific terms may align with emerging threats, this correlation is based on temporal trends and does not confirm causation. These observations provide valuable early signals but require further cross-validation with incident data.

The NMF model also decomposed the corpus into five topics, although the corresponding proportions per topic were more fixed than the proportions obtained from the LDA model, as shown in Fig. 5. Topic 4 is similar to Topic 3 in the LDA model, in which a specific communication tool (i.e., XMPP) and user status (i.e., prestige/reputation) are the focal points in the discussions identified in the XMPP-related threads. Topic 0 demonstrated slight deviations throughout, which indicates that both the trends focused on XMPP discussions and forum reputation remained largely the same during the examined period.

Topic 1 relates to the exchange of emails, which consists of posts and threads focused on email messaging, mainly on Yahoo or other conventional email services. This topic showed nearly no significant shifts, implying that the email conversations about that particular activity were steady over time.

Topic 2 concerns the BTC discussions, such as how long ago, in hours or days, the forum posts and threads revolved around a given definition. Similar to Topic 2 in the LDA model, this topic indicates the steadiness of conversations about cryptocurrencies. That is, this topic is constant to an extent, which means the discussions about BTC were also stable, with no real increase or decrease to observe.

Topic 3 was based on forum contacts and private messaging in which there was an emphasis on communications as well as hacking activities. Hence, it is similar to Topic 0 in the LDA model. There is an observable slow increase in Topic 3, which corresponds to the slight change in Topic 0 in the LDA model. This change indicates an increasing emphasis on private interactions within the forums.

Topic 4 describes the discussion of lessons focused on strategies, platforms, and user complaints, which are similarly associated with Topic 4 in the LDA model. This topic is also relatively stable, suggesting that discussions about the services offered to users and their rankings were present throughout the discourse, although not overwhelmingly so.

Compared with the LDA model, the NMF model provides a more balanced and consistent view of the proportions of topics, which suggests that the NMF model captures the underlying structure of the dataset without detecting sharp changes or emerging topics. The stability of the NMF model is useful for understanding general patterns over time but may be less effective at identifying rapidly changing trends or emerging threats. However, the NMF topics were closely aligned with the LDA topics in terms of their contents (e.g., discussions around XMPP, BTC, and user reputation), but the lack of significant changes in the proportions of topics means that the NMF model does not reveal dynamic shifts, as in the LDA model.

VI. CONCLUSION AND FUTURE WORK

Our research objective was to discover and classify potential CTI by gathering and analyzing data from surface web, deep web, and dark web sources in addition to well-known hacker forums. We developed ML and DL methods for classification based on several ML and neural network models. SVM with BOW (binary term weight) achieved the highest accuracy of 93.67 and 96.35 for binary and multiclass classification, respectively. Moreover, we used two topic modeling algorithms, namely, the LDA and NMF models, to extract the top topics from varying numbers of topics. In particular, the LDA model is well-suited for identifying emerging trends and is useful for real-time threat monitoring in cybersecurity, while the NMF model provides a more stable view of the topics, which means that the results are relatively consistent throughout the examined period. In addition, the NMF model is effective for capturing overall patterns; however, it may not be as sensitive to detecting shifts or emerging threats as the LDA model. The proposed framework provides dual capability of real-time classification and trend-based threat detection from surface, deep, and dark web sources. These address a real-time cybersecurity challenge on existing research. As future work, we aim to enhance the adaptability of the classification component by integrating automatic category discovery methods. We will use techniques such as clustering, neural embedding models, or unsupervised topic modeling to detect latent or evolving threat categories without relying on predefined labels. This approach will improve detection of unknown threats and allow the framework to remain effective in a dynamic cyber environment.

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ALEMAYEHU TILAHUN HAILE (Graduate Student Member, IEEE) received the master's degree in computer engineering from the Ethiopian Defense Engineering University in 2015. He is currently working toward the Ph.D. degree with the School of Information Technology and Engineering, Addis Ababa Institute of Technology. His research interests include cyber operations, including both cyber defense and offensive operations, with a specific focus on cybersecurity, cyber threat intelligence, and artificial intelligence.



SURAFEL LEMMA ABEBE received the Ph.D. degree from the International Doctoral School of the University of Trento, Italy, within the Software Engineering (SE) Research Unit at Fondazione Bruno Kessler (FBK). He was a postdoctoral fellow with the Software Analysis and Intelligence Lab (SAIL), School of Computing, Queen's University, Kingston, ON, Canada. He is currently an associate professor with the College of Technology and Built Environment, Addis Ababa University, Addis Ababa, Ethiopia. His research interests include ICT for development, application of artificial intelligence, program comprehension, software evolution, and source code and source code artifact analysis.



HENOCK MULUGETA MELAKU received the Ph.D. degree in computer engineering from Addis Ababa University (AAU), Addis Ababa, Ethiopia. He is currently an Assistant Professor with the School of Information Technology and Engineering (SITE), AAU. He has authored more than 18 research articles in networking and cybersecurity and has mentored more than 50 graduate students for more than 20 years. He is currently a postgraduate program coordinator for cybersecurity and AI programs. His expertise extends to collaborations

with IGAD's Security Sector Program (SSP), where he provides cybersecurity and cyber-intelligence training to law enforcement agencies in Ethiopia, Uganda, and Kenya, and affiliations with the Ethiopian Artificial Intelligence Institute (EAI). His research interests include cybersecurity and cyber intelligence, IoT, wireless ad hoc and sensor networks, and cybersecurity governance and risk management, supported by active memberships in professional associations like the Ethiopian Cybersecurity Association and the Ethiopian Society of Electrical Engineers.